

The LOPE Method: Improving Consistent Property Extraction for Scientific Knowledge Graphs Using LLMs

Sandra Schaftner

Martin Gaedke

sandra.schaftner@informatik.tu-chemnitz.de

martin.gaedke@informatik.tu-chemnitz.de

Chemnitz University of Technology

Chemnitz, Germany

Abstract

In the era of Generative AI, Scientific Knowledge Graphs (SKGs) have gained substantial importance as they provide a structured data foundation for fact grounding and scientific verification. They have become a cornerstone for Retrieval-Augmented Generation (RAG) and help detect and mitigate hallucinations of Large Language Models (LLMs), one of the major challenges for creating reliable outputs. However, creating comprehensive, content-rich SKGs remains a significant challenge, as current automated methods often fail to capture the semantic depth required to describe research contributions accurately. Conversely, manual crowdsourcing approaches are often time-consuming and error-prone, leading to semantic inconsistencies in the data.

To address these limitations, we present the LOPE (LLM-driven Ontology-based Property Extraction) method. Our automated approach advances LLM-based property extraction by combining semantically optimized prompting with a high-performance open-weight model and a vector-based ontology matching step. By aligning extracted terms to a standardized vocabulary, our solution improves semantic consistency compared to existing crowdsourcing approaches, thereby increasing the machine actionability and interoperability of the resulting SKG for downstream scientific analysis. The paper concludes with an evaluation using a validated LLM judge, demonstrating that LOPE highly significantly outperforms baseline methods.

CCS Concepts

• **Computing methodologies** → **Information extraction**; • **Information systems** → Digital libraries and archives; **Document representation**.

Keywords

Scientific Knowledge Graphs, Large Language Models, Relation Extraction, Relation Mapping, Open Research Knowledge Graph, Prompt Engineering

ACM Reference Format:

Sandra Schaftner and Martin Gaedke. 2026. The LOPE Method: Improving Consistent Property Extraction for Scientific Knowledge Graphs Using

LLMs. In *Companion Proceedings of the ACM Web Conference 2026 (WWW Companion '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3774905.3795079>

1 Introduction

The exponential growth of scientific publications has created an urgent need for machine-actionable knowledge representation. Scientific Knowledge Graphs (SKGs) have emerged as essential infrastructure to manage this flood [7, 42], serving as trusted foundations to ground AI applications, enhance Retrieval-Augmented Generation (RAG), and mitigate hallucinations of Large Language Models (LLMs) [27, 40].

Despite this potential, widespread adoption is hindered by SKG construction challenges. High-quality manual curation is too laborious to scale and crowdsourcing approaches often result in varying data quality [9, 23], whereas existing automated methods struggle to capture the necessary semantic depth [30]. This creates a dilemma where the demand for broad knowledge coverage conflicts with the strict requirements for data quality. Particularly, the task of Relation Extraction (RE) remains a critical hurdle. Most research operates in a closed setting, restricting extraction to predefined properties. Yet, capturing the full semantic breadth of research requires Open RE — the discovery of unrestricted properties [12]. However, Open RE creates a critical bottleneck: semantic heterogeneity. Without mapping the unrestricted, noisy outputs of LLMs to a standardized vocabulary, the resulting data remains disconnected strings rather than actionable knowledge, rendering it useless for downstream AI tasks or FAIR-compliant SKGs [29].

To address this, we present LOPE (LLM-driven Ontology-based Property Extraction). Unlike prior approaches that sacrifice flexibility for consistency (Closed RE) or consistency for flexibility (Open RE), LOPE bridges this gap. It employs a vector-based mapping mechanism to ground the unrestricted outputs of LLMs into a curated, standardized vocabulary. Our main contributions are:

- **High-Fidelity Extraction Basis:** We demonstrate that advanced prompt engineering is the necessary prerequisite for automated SKG construction, yielding highly significant improvements over baselines [23]. By instructing the model to strictly isolate schema properties from content values, we create the structural quality required for successful ontology matching.
- **Context and Model Optimization:** We perform a systematic analysis of context scope and model architecture. Contrary to common retrieval assumptions, we demonstrate that



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW Companion '26, Dubai, United Arab Emirates*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2308-7/2026/04

<https://doi.org/10.1145/3774905.3795079>

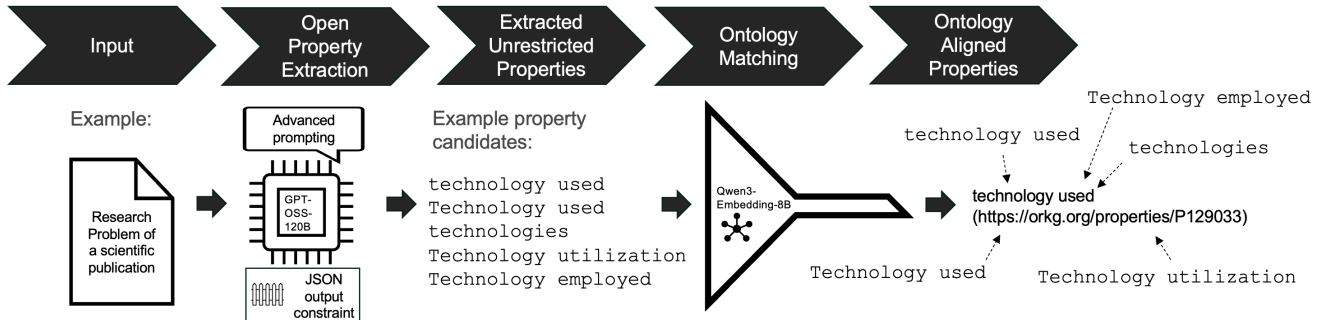


Figure 1: The LOPE Method. Overview of the pipeline for automated property extraction. (1) The input is the Research Problem description of a scientific publication. (2) Open Property Extraction utilizes a SOTA open-weight LLM with advanced prompting techniques and a strict JSON output constraint to generate (3) Unrestricted Property Candidates with heterogeneous surface forms (e.g., “technology used”, “Technologies utilization”). (4) Property Matching maps the extracted strings to the established ORKG ontology using a SOTA embedding model resulting in (5) Ontology Aligned Properties (e.g., P129033).

minimizing input context to the Research Problem significantly reduces noise compared to processing full abstracts. Furthermore, we show that a capable open-weight model (GPT-OSS-120B) outperforms proprietary baselines (GPT-3.5) highly significantly.

- **Semantic Ontology Alignment:** We introduce a vector-based property matching mechanism that maps unrestricted extractions to a controlled vocabulary. This acts as a consistency filter to resolve semantic heterogeneity and ensure the interoperability required for FAIR data principles [1, 29].
- **Automated Alignment Metric:** We propose this matching rate not only as a cleaning step but as a rigorous automated evaluation metric to quantify the structural fit of generated properties against established schemas, complementing traditional semantic evaluations.
- **Validated Evaluation:** Using a DeepSeek-V3 judge — validated for high alignment with human experts — we show that LOPE highly significantly outperforms prior approaches.

To our knowledge, this is the first study to reconcile the unrestricted discovery of SKG properties with the strict consistency required for data integration. We conclude that LOPE uniquely enables moving beyond the limitations of rigid, predefined lists without suffering from the semantic fragmentation of open generation.

2 State of the Art

2.1 Relation Extraction for SKGs

The task of extracting semantic triples from scientific texts falls into the broader domain of RE. Terminologically, distinct nuances exist: Kejriwal [13] distinguishes between object properties (relations between entities) and datatype properties (attributes/literals). In the specific context of this study, however, the terms *properties* and *predicates* are used synonymously to denote the semantic edges connecting a research contribution (e.g., a research problem) to specific values [23].

The advent of LLMs has fundamentally transformed RE capabilities, enabling the identification of complex relationships without

rigid pattern constraints [12]. Jiang et al. [12] classify LLM-based extraction as Generative RE (GRE) and distinguish between closed, semi-open, and open approaches. The majority of current research focuses on closed GRE, where the model extracts relations based on a strictly predefined list of allowed properties [7, 10, 16, 30–32, 34, 36, 41].

However, applying closed RE limits the semantic depth of SKGs, as it restricts extraction to a fixed schema. This limitation is evident in the Computer Science Knowledge Graph (CS-KG) [7], which is — besides the Open Research Knowledge Graph (ORKG) [2] — the only SKG capturing detailed research content such as methods, datasets, or metrics [15, 20]. While the CS-KG relies on a closed approach with a compact schema, the ORKG targets a deep semantic representation across diverse scientific domains. Consequently, automating the ORKG’s current crowdsourcing workflow necessitates an Open GRE approach to cover this semantic breadth, as a rigid schema would be insufficient [12].

While some recent studies like Chen et al. [5] apply Open GRE to scientific texts, they typically focus on ad-hoc graph generation without strict alignment to an established ontology. Consequently, these approaches lack the standardization required for global interoperability.

2.2 Benchmarking Open Property Extraction

To benchmark Open GRE specifically for the standardized environment of the ORKG, a dedicated evaluation framework is required. To the best of the authors’ knowledge, the only study providing such a framework is the work by Nechakhin et al. [23]. Unlike Chen et al. [5], Nechakhin et al. established a rigorous Gold Standard dataset comprising 153 manually curated ORKG contributions, allowing for objective comparative benchmarking.

Yet, their findings were notably pessimistic regarding prompt engineering: they concluded that prompting techniques yielded no significant improvements over simple zero-shot approaches and recommended fine-tuning as the necessary path forward. Furthermore, they identified a lack of alignment between LLM-generated dimensions and human expertise. Given that this study provides

the only expert-validated Gold Standard for this specific task [22], we adopt their dataset and evaluation protocol.

Existing approaches focus solely on the generation aspect. To our knowledge, no prior study has successfully implemented Open GRE followed by a dedicated vector-based ontology matching step. This study pioneers this hybrid pipeline, demonstrating that advanced prompt engineering achieves extraction fidelity comparable to human curation, while the subsequent matching layer establishes the strict semantic interoperability required for machine-actionable SKGs.

3 The LOPE Method

We present LOPE, a property extraction pipeline that integrates a SOTA open-weight LLM with vector-based grounding to automate the construction of SKGs with high standardization. The architecture follows a five-step process, illustrated in Figure 1.

3.1 Input: Research Problem Description

As shown in step (1) of the pipeline, we utilize the Research Problem description of a scientific publication as the input. While conventional RAG often maximizes context window usage, our ablation analysis (see Section 5) reveals that processing full abstracts introduces significant semantic drift.

3.2 Open Property Extraction

Step (2), the Open Property Extraction, utilizes gpt-oss-120b, a high-parameter open-weight LLM. At the time of writing, it ranks among the leading open-weight models on benchmarks like the LMArena [6]. Technically, it relies on an efficient Mixture-of-Experts (MoE) architecture [4, 11], which significantly reduces computational load by activating only a fraction of parameters during inference.

To ensure both extraction quality and machine actionability, we apply a *schema-constrained prompting* strategy utilizing advanced prompting techniques [25, 28], which also have been shown to be effective specifically for RE tasks [31, 32], and a strict JSON output constraint [26] (see Figure 2).

This process generates (3) Unrestricted Property Candidates. As illustrated in Figure 1, these candidates often represent synonymous concepts through varied surface forms (e.g., “technology used” vs. “Technology utilization”), necessitating subsequent alignment.

3.3 Property Matching

Matching properties to unique URIs is the next step in the KG construction process following property extraction, often referred to as “relation mapping” [8, 19]. To avoid confusion with the term “mappings” used by Nechakhin et al. [23] for comparing the number of similar properties to the Gold Standard, this study adopts the term “matching” for aligning properties to URIs, which is also used in the literature [19].

Through matching, the extracted properties — initially just words — are linked to unique URIs, giving them semantic meaning within the chosen ontology while contributing to consistency and unambiguity [8, 19, 20]. Property matching is also part of the manual approach employed by the ORKG. When creating a new contribution, users are encouraged to reuse properties already present in the

```

1 <system role>
You are an 2 Expert Scientific Knowledge Graph Curator. Your task is to structure
scientific text by extracting relevant property names (schema) to categorize
research papers.
</system role>
<instructions>
3. Extract the 3 PROPERTY NAME (e.g., "Study location", "Methodology"), NOT the
specific content or value.
4. Abstract specific details into their general category:
  - If text says "We used Python" -> Output "Programming language"
  - If text says "Germany" -> Output "Study location"
  - If text says "accuracy was 95%" -> Output "Evaluation metric"
5. Output between 4 and 8 distinct properties.
6. Respond ONLY 6 in valid JSON format with a single key 'properties'.
</instructions>
7 <examples>
<example>
  <input>Mapping of mineral-bound buddingtonite in hydrothermally altered
  rocks from Nevada</input>
  <output>{"properties": ["Data used", "Research problem", "Analysis type",
  "Reference"]}</output>
</example>
<example>
  <input>Sustainable supply chain optimization</input>
  <output>{"properties": ["Research problem", "Resolution methods",
  "Uncertainty factor", "Sustainability goal", "Objective function"]}</output>
</example>
</examples>
<task>
Analyze the following research problem and extract the schema properties as JSON.
</task>
<input_text>
{RESEARCH_PROBLEM_VARIABLE}
</input_text>

```

Figure 2: The LOPE prompt for extracting properties based on a research problem. The relevant improvements are highlighted in gray and numbered: (1) XML-structured prompting, (2) Persona description, (3) Clear and detailed instructions, (4) Output format constraint (JSON), and (5) Relevant few-shot examples exploiting majority label bias.

ORKG rather than creating new ones. However, many contributors appear not to follow this guideline, as browsing the ORKG properties reveals numerous duplicates. To address this issue and resolve the semantic heterogeneity of the generated candidates automatically, we implement a high-precision vector mapping mechanism.

Embedding Model Selection. We deploy the SOTA model Qwen3-Embedding-8B¹. The selection was strictly driven by its top performance on the Massive Text Embedding Benchmark (MTEB) [21], standing as the leading open-weight model in the relevant categories of *Semantic Textual Similarity (STS)* and *Clustering* at the time of writing.

Matching Process. The matching is performed via Nearest Neighbor Search in a local vector index (ChromaDB²) containing embeddings for the entire ORKG vocabulary (> 10,000 properties), which were retrieved via the ORKG REST API³. We apply a strict semantic distance threshold of $\delta < 0.15$ (cosine distance), empirically determined for the Qwen3-Embedding-8B embedding space. Candidates exceeding this distance are discarded as noise. This process results in (5) Ontology Aligned Properties, where raw strings are replaced by unique identifiers (e.g., P129033). This strict grounding prevents the proliferation of synonymous properties, ensuring that the resulting graph remains cohesive and directly queryable via the standardized ORKG vocabulary.

¹<https://huggingface.co/Qwen/Qwen3-Embedding-8B>

²<https://pypi.org/project/chromadb/>

³<https://tibhannover.gitlab.io/orkg/orkg-backend/api-doc/>

Table 1: Quantitative evaluation across all prompting strategies. The proposed method *GPT-OSS-120B (LOPE Prompt) achieves the best performance across all metrics evaluated by the validated DeepSeek-V3 judge.

Method	Evaluation by DeepSeek-V3		
	Alignment (1–5) ↑	Deviation (1–5) ↓	Mappings (Count) ↑
<i>SOTA</i>			
GPT-3.5 (Zero-Shot)	2.53	2.71	0.06
GPT-3.5 (Few-Shot)	2.78	2.43	0.12
GPT-3.5 (Chain-of-Thought)	2.85	2.07	0.15
<i>Prompting & Contextualization Experiments</i>			
GPT-3.5 (Combined)	2.93	2.25	0.22
GPT-3.5 (Improved Prompt)	3.24	1.95	0.59
GPT-3.5 (Improved + Abstract)	3.19	2.34	0.48
<i>LOPE</i>			
*GPT-OSS-120B (LOPE Prompt)	3.73	1.25	1.13
GPT-OSS-120B (LOPE + Abstract)	3.58	1.79	0.78

*The proposed LOPE method

4 Experimental Setup

To rigorously assess the performance of LOPE against established baselines, we adopt the evaluation protocol of the baseline study by Nechakhin et al. [23]. Our evaluation strategy is two-fold: First, we evaluate the generative extraction quality against the Gold Standard using a validated LLM-as-a-Judge approach. Second, we independently assess the interoperability by measuring the success rate of the subsequent vector-based ontology matching against the full ORKG vocabulary.

4.1 Dataset

We utilize the Gold Standard dataset established by Nechakhin et al. [22], comprising 153 scientific publications manually curated in the ORKG. Departing slightly from the original study, we reserved three entries (the first from each third of the dataset) for use as few-shot examples. Consequently, the final evaluation dataset contains 150 contributions.

4.2 Experimental Configurations

To provide a comprehensive analysis, we benchmark our approach against established methods and conduct systematic ablation studies. We categorize the experimental setups into three groups:

1. *SOTA Baselines (Nechakhin et al.)* We replicate the methods evaluated in the primary benchmark study [23] using the Research Problem as input. Since Nechakhin et al. did not specify the exact model version, we evaluated both gpt-3.5-turbo-0125 and gpt-3.5-turbo-1106. Consistent with prior studies [5, 32], we observed slight performance variances between versions. To ensure rigorous benchmarking, we report the results of the stronger version gpt-3.5-turbo-0125 as the baseline. The specific prompting strategies replicated from the baseline study are defined as follows:

- **GPT-3.5 (Zero-Shot):** A direct instruction approach that relies solely on the model’s pre-trained knowledge.

- **GPT-3.5 (Few-Shot):** Enables in-context learning by providing the model with the examples used in the baseline study.
- **GPT-3.5 (Chain-of-Thought):** Instructs the model to “think step-by-step” to induce intermediate reasoning.

Automated Execution & Guardrailing. To ensure a seamless, fully automated evaluation pipeline without human intervention while strictly preserving the original prompt structures from Nechakhin et al. (which lack a JSON output constraint), we implemented an output guardrailing mechanism [24] for these baselines. Although a regex-based post-processing step was used to isolate Python lists from the models’ verbose responses, it could not handle cases where list syntax was completely missing. To resolve this, a recursive repair strategy was introduced: if post-processing fails, the LLM is fed its own incorrect output along with the original task and instructed to correct the format. This approach demonstrated a 100% success rate. The recovery mechanism was required 5 times for gpt-3.5-turbo-0125 and 2 times for gpt-3.5-turbo-1106, enabling the processing of the entire dataset in a single run.

2. *Prompting & Contextualization Experiments.* To isolate the effects of our prompting strategy and context selection, we evaluate three intermediate configurations using the same GPT-3.5 architecture:

- **GPT-3.5 (Combined):** Merges Few-Shot and CoT strategies.
- **GPT-3.5 (Improved Prompt):** Applies our advanced LOPE prompt to the proprietary model.
- **GPT-3.5 (Improved + Abstract):** Uses the improved prompt but processes the full abstract instead of the Research Problem.

3. *The LOPE Method (Proposed Approach).* Finally, we evaluate our proposed configuration utilizing the open-weight MoE model gpt-oss-120b:

- ***GPT-OSS-120B (LOPE Prompt):** The core LOPE method using the schema-constrained, advanced prompt on the Research Problem.
- **GPT-OSS-120B (LOPE + Abstract):** The context ablation variant processing the full abstract.

4.3 Evaluation of Extraction Quality

The evaluation of LLM results follows the methodology of Nechakhin et al. [23] to ensure comparability. We consider this method effective as it avoids rigid exact-matching against the Gold Standard [22]. Instead, it employs a “soft” evaluation based on semantic alignment and deviation. This approach is increasingly supported by literature advocating for flexible evaluation metrics in open extraction tasks [12, 31, 36].

LLM-as-a-Judge Validation. Following Nechakhin et al., we employ an LLM-as-a-Judge to automate the scoring. To select the most reliable judge, we conducted a comparative validation study benchmarking DeepSeek-V3 against a set of strong competitors — Qwen3-30B-A3B, Mistral-Small-3.1, GLM-4-32B, Gemma-3-27B,

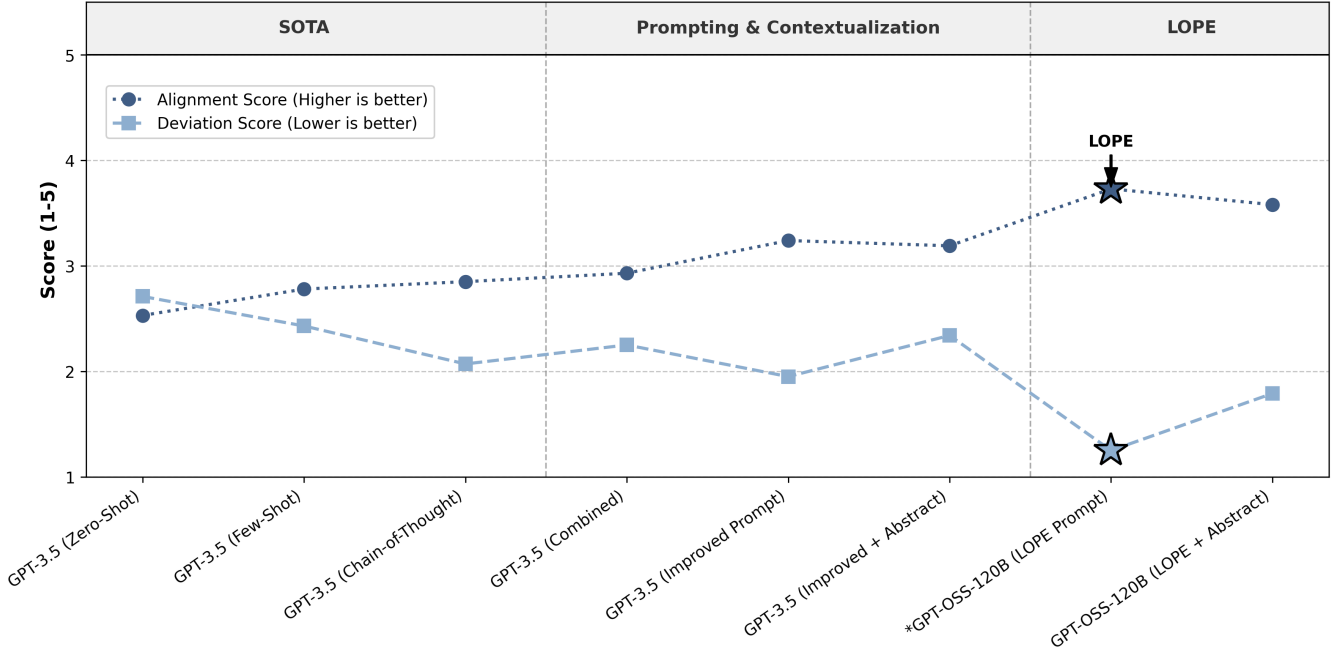


Figure 3: Performance trajectory of alignment vs. deviation. The proposed LOPE method (marked with stars) maximizes semantic alignment while minimizing deviation from the Gold Standard dataset. [22]

GPT-4.1, and Gemini-2.0-flash — yielding the result that DeepSeek-V3 exhibits the highest alignment with human expert judgment.⁴ Based on this validation, DeepSeek-V3 rates the extracted properties on the metrics defined by Nechakhin et al.:

- (1) **Alignment Score (1–5):** Measures the semantic similarity between the extracted candidate and the Gold Standard property. A score of 5 indicates a perfect synonym or exact match.
- (2) **Deviation Score (1–5):** Quantifies the degree of semantic error. A high deviation score (5) indicates that the extracted term is unrelated to the Gold Standard property.
- (3) **Mappings (Count):** Quantifies the volume of valid extractions. It represents the absolute number of extracted candidates that the Judge identifies as semantically equivalent to a property in the Gold Standard list.

4.4 Evaluation of Interoperability (Matching)

In addition to the generative quality, we assess the system’s ability to create interoperable data using the Matching Rate. Similar to Nechakhin et al. [23], who employed embedding-based metrics (SciNCL) to compare generations against the Gold Standard, we utilize vector embeddings generated by Qwen3-Embedding-8B to assess semantic proximity. However, rather than calculating an average similarity score, we employ embeddings as a constructive step to link candidates to the entire ORKG vocabulary. The Matching Rate measures the proportion of extracted candidates that are

successfully mapped to a valid ORKG property URI within a defined semantic distance threshold ($\delta < 0.15$). This metric directly quantifies the volume of machine-actionable data generated by the pipeline.

Statistical Analysis. To ensure the reliability of our findings, we applied rigorous statistical testing. For the comparative evaluation of generation quality, we assessed significance using the Binomial Sign Test on the pairwise judgments. For categorical comparisons (Matching Rates), we utilized Pearson’s Chi-Square test. All reported improvements are statistically significant at $p < 0.05$ or highly significant at $p < 0.001$.

5 Results and Discussion

This section presents the quantitative findings of our evaluation, focusing on the impact of prompt engineering, model capabilities, and context selection. We analyze the performance improvements in terms of semantic alignment and deviation, followed by an assessment of the system’s interoperability via ontology matching⁵.

5.1 Impact of Prompt Engineering

The quantitative results (see Table 1) challenge the baseline assumption that prompt engineering is ineffective for Open GRE in scientific domains, as previously suggested by Nechakhin et al. [23].

⁴The scripts, input files, and results of this comparative validation study are available in the `llm-judge-validation` folder at <https://w3id.org/allmac/lope/>.

⁵The complete results and evaluation data, including all statistical reports, bar charts and significance heatmaps, as well as the complete script and data is available at <https://w3id.org/allmac/lope/>.

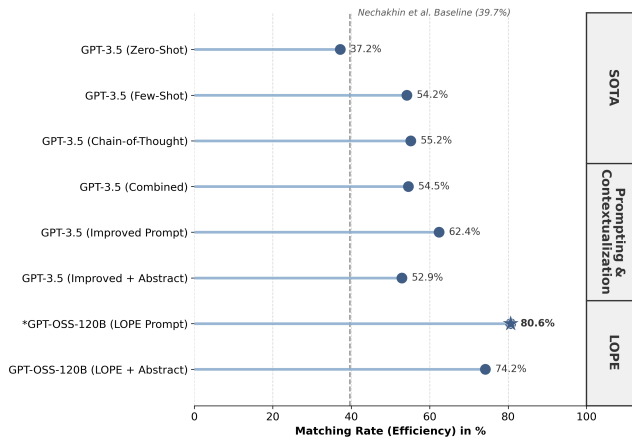


Figure 4: Efficiency comparison (Matching Rate). The proposed LOPE method (*GPT-OSS-120B) achieves an 80.6% matching rate, highly significantly outperforming the Nechakhin et al. Baseline (39.7%, dashed line) and demonstrating the highest efficiency among all methods.

SOTA vs. Experimental Configurations. First, we analyze the transition from the *SOTA Baselines* to our *Prompting Experiments*. Comparing the GPT-3.5 (Zero-Shot) baseline against our GPT-3.5 (Improved Prompt) on the same architecture and context reveals highly significant gains. Specifically, the Improved Prompt increased the Alignment Score from 2.53 to 3.24 and reduced the Deviation Score from 2.71 to 1.95. This distinct improvement in both dimensions is visually presented in Figure 3.

Interestingly, our intermediate GPT-3.5 (Combined) configuration yielded unexpected results. While previous studies in RE suggested that combining Few-Shot and Chain-of-Thought strategies yields synergistic effects [31], we observed a performance plateau where stacking these techniques provided no tangible benefit over the standalone baselines. Instead, these results indicate that the primary bottleneck in previous studies was the lack of structural guidance. By strictly enforcing the separation between schema (properties) and instance (values), the model shifts from extracting loose concepts to valid property candidates. A significant contribution is certainly also made by the specific prompting strategies employed, whose effects have been proven in other studies [3, 35], also specifically for RE [31, 32], as well as the specific selection of examples [14, 17, 18].

Impact of Example Selection. A key factor in our prompt’s success is its exploitation of the Majority Label Bias, defined by Zhao et al. [39] as a bias “towards answers that are frequent in the prompt.” An analysis of the Gold Standard dataset [22] reveals that every single one of the 153 property lists contains the property *research problem*. This ubiquity stems from modeling best practices⁶ established by the ORKG, which guide users on how to properly define and label new graph elements, and legacy UI designs where this field

⁶https://orkg.org/help-center/article/42/Modeling_best_practices_for_resources_and_properties (Accessed 30 Nov 2025)

Table 2: Detailed analysis of the vector-based property matching. The upper section displays exemplary Accepted Matches (Distance < 0.15), demonstrating the system’s ability to consolidate semantic synonyms into canonical ORKG properties. The lower section displays exemplary Rejected Candidates (Distance ≥ 0.15), illustrating how the threshold effectively filters out unrelated or overly vague terms.

Extracted Candidate	Mapped ORKG Property	Dist.
<i>Accepted Matches (Semantic Consolidation)</i>		
Geographic scope	geographical scope	0.01
Research method	research methodology	0.04
Data source	source of data	0.06
Material used	material	0.08
Methodological approach	methodology	0.08
Evaluation approach	Evaluation methods	0.09
Analysis type	Type of analysis	0.10
Research domain	has research domain	0.10
Mitigation approach	mitigation strategies	0.10
Features used	used features	0.10
Reference database	Reference data	0.14
Feature set	Features	0.14
Theoretical framework	Theoretical background	0.14
Implementation approach	has implementation approach	0.15
Stakeholder group	Stakeholders	0.15
<i>Rejected Candidates (Noise Filtering)</i>		
Target disease	disease name	0.15
Ontology scope	Ontology	0.15
Target compound	Lead compound	0.15
Task definition	Task name	0.16
Study context	study	0.16
Normalization technique	Normalization types	0.17
Proposed framework	Proposed model	0.18
Recommender system	User recommendation	0.19

was pre-configured as a default field⁷. Consequently, our optimized prompt strategically included three examples (randomly selected from the dataset) that all contained *research problem*. This alignment with the dataset’s inherent bias led the LLM to consistently extract this key property, whereas the SOTA Few-Shot examples lacked this alignment, leading to lower recall.

5.2 Ablation Study: Context and Model

Having established the optimal prompt, we compare the *Experimental Configurations* against the final *LOPE Method*.

Context Selection. Contrary to prior research suggesting that expanding context generally enhances LLM performance [14, 33, 38], our results show that processing the full abstract consistently degrades performance compared to the focused Research Problem. This negative impact is observed across both model architectures:

- For GPT-3.5, adding the abstract worsened the Deviation Score from 1.95 to 2.34.

⁷As can be seen in an earlier demonstration video: https://youtu.be/8rqkQe_2lUE?si=QdhIPdU7AzEcT6k2 (Accessed 30 Nov 2025)

- For **GPT-OSS-120B**, the same pattern occurred, increasing deviation from 1.25 to 1.79.

This suggests that full abstracts introduce semantic noise that leads the model to generate candidates that are overly specific to the individual paper. This contradicts the design principle of SKG properties, which aim for reusability across contributions rather than capturing granular, content-specific details.

Model Capability (The LOPE Advantage). The transition from GPT-3.5 to the open-weight GPT-OSS-120B yielded a further substantial performance leap, comparable in magnitude to the gain achieved by advanced prompt engineering. As shown in Table 1, the proposed LOPE Method (GPT-OSS-120B with Improved Prompt) achieved the highest Alignment (3.73) and the lowest Deviation (1.25) of all tested setups. Notably, LOPE highly significantly ($p < 0.001$) outperforms both the SOTA Baselines and the best GPT-3.5 configuration (utilizing the identical advanced prompt), demonstrating that modern MoE architectures, combined with precise instruction tuning, establish a new state-of-the-art for this task.

5.3 Interoperability through Ontology Matching

Beyond generative quality, the pivotal contribution of LOPE is the creation of machine-actionable data. The results for the Matching Rate highlight the efficiency of the vector-based alignment layer (see Figure 4):

- **SOTA Baseline (Nechakhin et al.):** Only 39.7% of extracted candidates (when processed through our matching step) could be mapped to existing ORKG URIs.
- **LOPE Method (GPT-OSS-120B):** Achieved a Matching Rate of 80.6%, successfully linking the vast majority of extracted properties to the ontology.

This twofold increase demonstrates that our pipeline effectively bridges the gap between open-ended generation and controlled vocabularies.

Table 2 provides a detailed analysis of this mechanism, utilizing exemplary matches drawn from diverse publications within the dataset. The upper section illustrates how LOPE effectively consolidates synonymous surface forms into canonical ORKG properties. Conversely, the lower section demonstrates the effectiveness of the threshold ($\delta < 0.15$) in filtering out noise. This strictly grounded approach ensures that the resulting KG remains cohesive and directly queryable, while simultaneously flagging potential novel properties for human review.

6 Learnings and Future Work

While our results demonstrate the effectiveness of the LOPE pipeline, we identify specific challenges regarding the underlying ontology and the broader scope of SKG construction that open avenues for future research.

Ontology Redundancy. A significant challenge in the current ORKG infrastructure is ontology redundancy. For example, searching for the property “method” returns more than 20 IDs representing the same concept,⁸ without considering plural forms like “methods”, compound terms like “used method”, or related words such as

“methodology”. However, a crucial advantage of our vector-based approach is its deterministic nature. By consistently retrieving only the single nearest neighbor, the mapping for any given input string remains constant. This effectively mitigates the issue of duplicate IDs within the ORKG: even if multiple URIs exist for the label “method”, the embedding-based pipeline will consistently map the string “method” to the exact same URI every time. This creates a standardizing effect, ignoring redundant IDs and enforcing consistency across the automated pipeline, which is essential for subsequent machine-actionable tasks.

Ontology Consolidation and Evolution. Nevertheless, to fully leverage the resulting advantages such as consistency and unambiguity, the current list of over 10,000 properties in the ORKG would need to be consolidated in a real-world application, which constitutes a significant undertaking [37]. For cases where multiple semantically identical properties exist, the consolidation process should select the property that best aligns with the modeling best practices (e.g., singular, short labels) that the ORKG team provides to the contributors and replace others. Furthermore, future iterations of LOPE could integrate a mechanism to allow LLM agents to propose new properties for the ontology when no suitable match is found, thereby also enabling the coverage of specific concepts from long-tail research domains. These newly proposed properties would require approval from human experts to ensure quality standards. This approval process is expected to be far less labor-intensive than the current manual extraction of all properties from research publications.

Towards a Complete SKG Construction Framework. Finally, when designing a comprehensive SKG construction pipeline, it is crucial to address inherent LLM issues such as hallucinations and non-determinism. Jiang et al. [12] proposed a framework for factualness checks to mitigate these challenges. In the current use case of Open GRE without Entity Extraction, distinguishing between creativity and hallucination remains challenging. However, with the inclusion of Entity Extraction and Mapping to standard vocabularies (e.g., DBpedia, Wikidata), the full triples could be more effectively evaluated for factualness. Consequently, the guardrail mechanism could be extended to encompass semantic content verification and fact-checking protocols [24]. The development of such a complete framework for extracting full triples, incorporating these improvements, is currently underway by the authors.

7 Conclusion

This study presented LOPE, a novel pipeline reconciling the flexibility of Open GRE with the strict standardization required for SKGs. By combining schema-constrained prompting with open-weight models, we achieved extraction fidelities that highly significantly outperform established baselines.

Crucially, the integration of a dedicated vector-based matching layer enhances the process by directly addressing the issues of inconsistency and fragmentation inherent in current SKGs. This matching step improves result quality by ensuring that extracted candidates align with a predefined ontology, allowing for criteria such as generalizability and interoperability to be systematically enforced. Consequently, LOPE transforms raw LLM outputs into

⁸<https://orkg.org/search?q=method&type=Predicate> (Accessed 30 Nov 2025)

a cohesive, machine-actionable SKG, paving the way for scalable automated curation.

Declaration on Generative AI

During the preparation of this work, the authors utilized Generative AI for translation, rewording, improving style, and condensing text. Furthermore, AI assistance was used to generate Python code for plotting data. The authors reviewed and edited the content as needed and take full responsibility for the publication's content.

Acknowledgments

The authors gratefully acknowledge Nechakhin, D'Souza, and Eger for publishing the dataset [22], prompts, and experimental setup. This made it possible to replicate, understand, and build upon their study. We hope that through our ideas and findings presented in this paper, as well as our publicly available code, prompts, and result files on GitHub (<https://w3id.org/allmac/lope>), we can contribute back to Nechakhin, D'Souza, and Eger, and to their project.

This work is supported by the European Union's HORIZON Research and Innovation Programme under grant agreement No 101120657, project ENFIELD (European Lighthouse to Manifest Trustworthy and Green AI).

References

- [1] Sören Auer, Vinodh Ilango, Markus Stocker, Sanju Tiwari, and Lars Vogt (Eds.). 2024. *Open Research Knowledge Graph*. Cuvillier Verlag, Göttingen. doi:10.34657/13789
- [2] Sören Auer, Allard Oelen, Muhammad Haris, Markus Stocker, Jennifer D'Souza, Khair Eddine Farfar, Lars Vogt, Manuel Prinz, Vitalis Wiens, and Mohamad Yaser Jaradeh. 2020. Improving Access to Scientific Literature with Knowledge Graphs. *Bibliothek Forschung und Praxis* 44, 3 (2020), 516–529. doi:10.1515/bfp-2020-2042
- [3] Tom B. Brown et al. 2020. Language Models are Few-Shot Learners. *CoRR* abs/2005.14165 (2020). arXiv:2005.14165
- [4] Weilin Cai et al. 2025. A Survey on Mixture of Experts in Large Language Models. *IEEE Transactions on Knowledge and Data Engineering* 37, 7 (2025), 3896–3915. doi:10.1109/TKDE.2025.3554028
- [5] Haoting Chen et al. 2025. Open Local Knowledge Graph Construction from Academic Papers Using Generative Large Language Models. In *Companion Proceedings of the ACM on Web Conference 2025 (WWW '25)*. Association for Computing Machinery, New York, NY, USA, 2551–2559. doi:10.1145/3701716.3717820
- [6] Wei-Lin Chiang et al. 2024. Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. *arXiv:2403.04132* (2024). Leaderboard: <https://lmarena.ai>.
- [7] Danilo Dessi et al. 2025. CS-KG 2.0: A Large-scale Knowledge Graph of Computer Science. *Scientific Data* 12, 1 (June 2025), 964. doi:10.1038/s41597-025-05200-8
- [8] Danilo Dessi et al. 2022. SCICERO: A deep learning and NLP approach for generating scientific knowledge graphs in the computer science domain. *Knowledge-Based Systems* 258 (2022), 109945. doi:10.1016/j.knsys.2022.109945
- [9] Jennifer D'Souza et al. 2024. Quality Assessment of Research Comparisons in the Open Research Knowledge Graph: A Case Study. *JLIS.it* 15, 1 (Jan. 2024), 126–143. doi:10.36253/jlis.it-547
- [10] Suramya Jadhav et al. 2025. Leveraging Large Language Models for Biomedical Knowledge Graph Construction and Querying: An Advanced NLP Approach. In *Companion Proceedings of the ACM on Web Conference 2025 (WWW '25)*. Association for Computing Machinery, New York, NY, USA, 2560–2566. doi:10.1145/3701716.3717817
- [11] Albert Q. Jiang et al. 2024. Mixtral of Experts. arXiv:2401.04088 [cs.LG]
- [12] Pengcheng Jiang et al. 2024. GenRES: Rethinking Evaluation for Generative Relation Extraction in the Era of Large Language Models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*. Association for Computational Linguistics, 2820–2837. doi:10.18653/v1/2024.naacl-long.155
- [13] Mayank Kejriwal. 2022. Knowledge Graphs: A Practical Review of the Research Landscape. *Information* 13, 4 (2022). doi:10.3390/info13040161
- [14] Vamsi Krishna Kommineni, Birgitta König-Ries, and Sheeba Samuel. 2024. From human experts to machines: An LLM supported approach to ontology and knowledge graph construction. arXiv:2403.08345 [cs.CL] <https://arxiv.org/abs/2403.08345>
- [15] Filip Kovačević, Alaa El-Ebshihy, Florina Piroi, and Andreas Rauber. 2024. Extending Content-based Scientific Knowledge Graphs with Research Results. In *Proceedings of the Austrian Symposium on AI, Robotics, and Vision (AIROV 2024)*. Innsbruck University Press, Innsbruck, Austria. doi:10.15203/99106-150-2-16
- [16] Guozheng Li, Peng Wang, and Wenjun Ke. 2023. Revisiting Large Language Models as Zero-shot Relation Extractors. arXiv:2310.05028 [cs.AI] <https://arxiv.org/abs/2310.05028>
- [17] Jinyuan Li et al. 2023. Prompting ChatGPT in MNER: Enhanced Multimodal Named Entity Recognition with Auxiliary Refined Knowledge. arXiv:2305.12212 [cs.CL] <https://arxiv.org/abs/2305.12212>
- [18] Jiachang Liu et al. 2022. What Makes Good In-Context Examples for GPT-3?. In *Proceedings of Deep Learning Inside Out (DeeLIO 2022)*. Association for Computational Linguistics, 100–114. doi:10.18653/v1/2022.deeLIO-1.10
- [19] Jose L. Martinez-Rodriguez et al. 2018. OpenIE-based approach for Knowledge Graph construction from text. *Expert Systems with Applications* 113 (2018), 339–355. doi:10.1016/j.eswa.2018.07.017
- [20] Antonello Meloni et al. 2023. Integrating Conversational Agents and Knowledge Graphs Within the Scholarly Domain. *IEEE Access* 11 (2023), 22468–22489. doi:10.1109/ACCESS.2023.3253388
- [21] Niklas Muennighoff et al. 2023. MTEB: Massive Text Embedding Benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. 2002–2025. doi:10.18653/v1/2023.eacl-main.148
- [22] Vladyslav Nechakhin and Jennifer D'Souza. 2024. ORKG Properties and LLM-Generated Research Dimensions Evaluation Dataset. doi:10.25835/6OYN9D1N
- [23] Vladyslav Nechakhin, Jennifer D'Souza, and Steffen Eger. 2024. Evaluating Large Language Models for Structured Science Summarization in the Open Research Knowledge Graph. *Information* 15, 6 (2024). doi:10.3390/info15060328
- [24] OpenAI. 2024. *How to implement LLM guardrails | OpenAI Cookbook*. Retrieved November 30, 2025 from https://cookbook.openai.com/examples/how_to_use_guardrails
- [25] OpenAI. 2024. *Prompt engineering | OpenAI API*. Retrieved November 30, 2025 from <https://platform.openai.com/docs/guides/prompt-engineering>
- [26] OpenAI. 2025. Structured model outputs. <https://platform.openai.com/docs/guides/structured-outputs>. Accessed: 2025-12-22.
- [27] Shirui Pan et al. 2024. Unifying Large Language Models and Knowledge Graphs: A Roadmap. *IEEE Transactions on Knowledge and Data Engineering* 36, 7 (2024), 3580–3599. doi:10.1109/TKDE.2024.3352100
- [28] Elvis Saravia. 2022. *Prompt Engineering Guide*. <https://github.com/dair-ai/Prompt-Engineering-Guide> DAIR.AI.
- [29] Markus Stocker et al. 2023. FAIR scientific information with the Open Research Knowledge Graph. *FAIR Connect* 1, 1 (2023), 19–21. doi:10.3233/FC-221513
- [30] Shilpa Verma et al. 2023. Scholarly knowledge graphs through structuring scholarly communication: a review. *Complex & Intelligent Systems* 9, 1 (Feb. 2023), 1059–1095. doi:10.1007/s40747-022-00806-6
- [31] Somn Wadhwa, Silvio Amir, and Byron Wallace. 2023. Revisiting Relation Extraction in the era of Large Language Models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 15566–15589. doi:10.18653/v1/2023.acl-long.868
- [32] Zhen Wan et al. 2023. GPT-RE: In-context Learning for Relation Extraction using Large Language Models. arXiv:2305.02105 [cs.CL] <https://arxiv.org/abs/2305.02105>
- [33] Xinyu Wang et al. 2022. Improving Named Entity Recognition by External Context Retrieving and Cooperative Learning. arXiv:2105.03654 [cs.CL] <https://arxiv.org/abs/2105.03654>
- [34] Xiang Wei et al. 2024. ChatIE: Zero-Shot Information Extraction via Chatting with ChatGPT. arXiv:2302.10205 [cs.CL] <https://arxiv.org/abs/2302.10205>
- [35] Lilian Weng. 2023. *Prompt Engineering*. Retrieved November 30, 2025 from <https://lilianweng.github.io/posts/2023-03-15-prompt-engineering/>
- [36] Rui Yang et al. 2025. Graphusion: A RAG Framework for Scientific Knowledge Graph Construction with a Global Perspective. In *Companion Proceedings of the ACM on Web Conference 2025 (WWW '25)*. Association for Computing Machinery, New York, NY, USA, 2579–2588. doi:10.1145/3701716.3717821
- [37] Antonio Zaitoun et al. 2023. OntoEval: an Automated Ontology Evaluation System. In *Companion Proceedings of the ACM Web Conference 2023 (WWW '23 Companion)*. ACM, New York, NY, USA, 82–85. doi:10.1145/3543873.3587318
- [38] Xin Zhang et al. 2022. Domain-Specific NER via Retrieving Correlated Samples. arXiv:2208.12995 [cs.CL]
- [39] Tony Z. Zhao et al. 2021. Calibrate Before Use: Improving Few-Shot Performance of Language Models. arXiv:2102.09690 [cs.CL]
- [40] Wayne Xin Zhao et al. 2025. A Survey of Large Language Models. arXiv:2303.18223 [cs.CL]
- [41] Yuqi Zhu et al. 2024. LLMs for Knowledge Graph Construction and Reasoning: Recent Capabilities and Future Opportunities. arXiv:2305.13168 [cs.CL]
- [42] Matthäus Zloch et al. 2025. Research Knowledge Graphs: The Shifting Paradigm of Scholarly Information Representation. In *The Semantic Web*. Springer Nature Switzerland, Cham, 140–154. doi:10.1007/978-3-031-94578-6_9